The empirical recurrence for OEIS A236723: recovering a conjecture that is not written down

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Abstract

OEIS A236723 records "Empirical recurrence of order 81", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 219$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 81, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 81 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A236723 is "Number of (n+1)X(3+1) 0..2 arrays with the maximum plus the upper median plus the lower median minus the minimum of every 2X2 subblock equal.". It has offset 1 and begins

$$550, 1956, 6879, 26724, 105206, 437291, 1848980, 8110850, \ldots$$

The entry states:

> Empirical recurrence of order 81 (see link above)

As of the "Last modified" line on the live entry (Jul 23 2025 09:29:28, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 219$ vertices, $a(n) = \iota^{\top} M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 219.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 438$ exact terms of the model returns order 81 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{81} c_i\, a(n-i),$$

c_1	19	c_{22}	8193668760892	c_{43}	25074011154440649847	c_{64}	−598272005100510
c_2	−20	c_{23}	2856949817013894	c_{44}	−153129807031252098372	c_{65}	2960557104830516
c_3	−1798	c_{24}	−1219503482717994	c_{45}	−15843684770378967428	c_{66}	1167050262603476
c_4	9450	c_{25}	−15721728148048404	c_{46}	189739591301992927068	c_{67}	−62094332631259
c_5	69144	c_{26}	11970202404374191	c_{47}	−1263273817670156856	c_{68}	−17629250716858
c_6	−626222	c_{27}	72490118340617011	c_{48}	−200473461472281892857	c_{69}	101430166920304
c_7	−1147293	c_{28}	−76387698530604466	c_{49}	19672586745808045007	c_{70}	19918124284037
c_8	22363651	c_{29}	−281582044490730208	c_{50}	180178292001806021449	c_{71}	−12589623362748
c_9	−7051715	c_{30}	373457520939873656	c_{51}	−31507469288111648903	c_{72}	−1594496650067
c_{10}	−525668926	c_{31}	924144500129542554	c_{52}	−137285912981659748548	c_{73}	11455928948480
c_{11}	850731759	c_{32}	−1476805579934470528	c_{53}	32927162432645667182	c_{74}	8214633344512
c_{12}	8811713480	c_{33}	−2563799028203664601	c_{54}	88293401520963497398	c_{75}	−723956452403
c_{13}	−23572348199	c_{34}	4843801468139216164	c_{55}	−26082361930669943375	c_{76}	−217645979136
c_{14}	−109263622261	c_{35}	5998389681223165985	c_{56}	−47668819076134529655	c_{77}	291154489344
c_{15}	413151822876	c_{36}	−13358256848414216370	c_{57}	16374154593743998394	c_{78}	581984256
c_{16}	1009691812464	c_{37}	−11763950053594776047	c_{58}	21459033183481535682	c_{79}	−6370258944
c_{17}	−5320455974297	c_{38}	31221771562274856361	c_{59}	−8279480719048483294	c_{80}	61341696
c_{18}	−6726021386460	c_{39}	19103438056050731142	c_{60}	−7987735820495146290	c_{81}	53673984
c_{19}	53408095907035	c_{40}	−62132189363878055564	c_{61}	3386826453947337800		
c_{20}	27397721975096	c_{41}	−25058908853032213293	c_{62}	2432907053122192808		
c_{21}	−431304925465946	c_{42}	105533695656173333564	c_{63}	−1118398662881837064		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 81, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $81 + 4$ terms removed returns order 81 as well, so $\deg q_{\min} = 81 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 20 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $81+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 81 that this sequence satisfies — and that family turns out to have exactly one member.

References

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